Simulate and Forecast a VEC Model

This example shows how to generate forecasts from a VEC(*q*) model several ways:

Open Script

- Monte Carlo forecast paths using the VEC(*q*) model directly
- Minimum mean square error (MMSE) forecasts using the VAR model representation of the VEC(*q*) model.
- Monte Carlo forecast paths using the VAR(*p*) model representation of the VEC(*q*) model.

This example follows from Estimate VEC Model Parameters Using egcitest.

Load Data and Preprocess

Load the `Data_Canada` data set. Extract the interest rate data.

```
load Data_Canada
Y = Data(:,3:end);
```

Estimate VEC(*q*) Model

Assuming that the interest rate data follows a VEC(2) model, fit the model to the data.

```
[~,~,~,~,reg] = egcitest(Y,'test','t2');
c0 = reg.coeff(1);
b = reg.coeff(2:3);
beta = [1; -b];
q = 2;
[numObs,numDims] = size(Y);
tBase = (q+2):numObs;               % Commensurate time base, all lags
T = length(tBase);                  % Effective sample size
DeltaYLags = zeros(T,(q+1)*numDims);
YLags = lagmatrix(Y,0:(q+1));       % Y(t-k) on observed time base
LY = YLags(tBase,(numDims+1):2*numDims);
for k = 1:(q+1)
    DeltaYLags(:,((k-1)*numDims+1):k*numDims) = ...
               YLags(tBase,((k-1)*numDims+1):k*numDims) ...
             - YLags(tBase,(k*numDims+1):(k+1)*numDims);
end
DY = DeltaYLags(:,1:numDims);       % (1-L)Y(t)
DLY = DeltaYLags(:,(numDims+1):end); % [(1-L)Y(t-1),...,(1-L)Y(t-q)]
X = [(LY*beta-c0),DLY,ones(T,1)];
P = (X\DY)';                        % [alpha,B1,...,Bq,c1]
alpha = P(:,1);
C = alpha*beta';                    % Error-correction coefficient matrix
B1 = P(:,2:4);                      % VEC(2) model coefficient
B2 = P(:,5:7);                      % VEC(2) model coefficient
c1 = P(:,end);
b = (alpha*c0 + c1)';               % VEC(2) model constant offset
res = DY-X*P';
EstCov = cov(res);
```

Monte Carlo Forecasts Using VEC Model

Specify a 10-period forecast horizon. Set `numPaths` to generate 1000 paths. Because *q*, the degree of the VEC model, is 2, reserve three presample observations for y_t.

```
forecastHorizon = 10;
numPaths        = 1000;
psSize          = 3;

PSY = repmat(Y(end-(psSize-1):end,:),1,1,numPaths); % Presample
YSimVEC = zeros(forecastHorizon,numDims,numPaths);  % Preallocate forecasts
YSimVEC = [PSY; YSimVEC];
```

Generate Monte Carlo forecasts by adding the simulated innovations to the estimated VEC(2) model.

```
rng('default'); % For reproducibility
for p = 1:numPaths
    for t = (1:forecastHorizon) + psSize;
        eps = mvnrnd([0 0 0],EstCov);
        YSimVEC(t,:,p) = YSimVEC(t-1,:,p) + (C*YSimVEC(t-1,:,p)')'...
                     + (YSimVEC(t-1,:,p) - YSimVEC(t - 2,:,p))*B1'...
                     + (YSimVEC(t-2,:,p) - YSimVEC(t - 3,:,p))*B2'...
                     + b + eps;
    end
end
```

`YSimVEC` is a 13-by-3-by-1000 numeric array. Its rows correspond to presample and forecast periods, columns correspond to the time series, and the pages correspond to a draw.

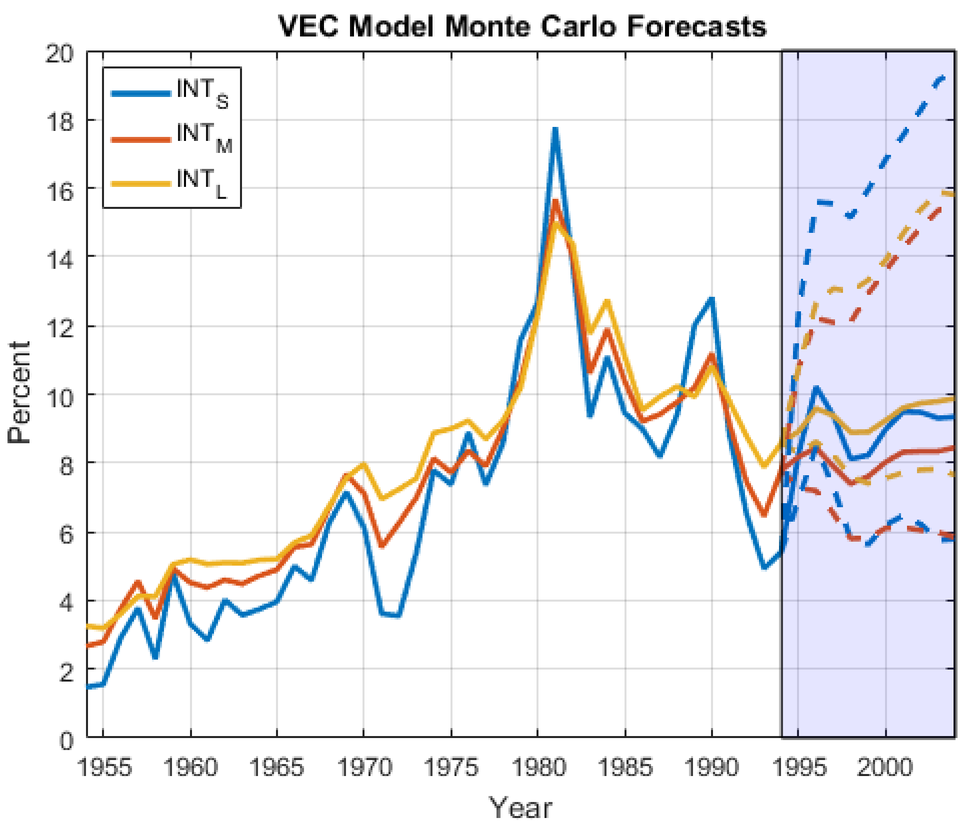
Compute the mean of the forecasts for each period and time series over all paths. Construct 95% percentile forecast intervals for each period and time series.

```
FMCVEC   = mean(YSimVEC((psSize + 1):end,:,:),3);
CIMCVEC = quantile(YSimVEC((psSize + 1):end,:,:),[0.25,0.975],3);
```

`FMCVEC` is a 10-by-3 numeric matrix containing the Monte Carlo forecasts for each period (row) and time series (column). `CIMCVEC` is a 10-by-3-by-2 numeric array containing the 2.5% and 97.5% percentiles (pages) of the draws for each period (row) and time series (column).

Plot the effective-sample observations, the mean forecasts, and the 95% percentile confidence intervals.

```
fDates = dates(end) + (0:forecastHorizon)';
figure;
h1 = plot([dates; fDates(2:end)],[Y; FMCVEC],'LineWidth',2);
h2 = gca;
hold on
h3 = plot(repmat(fDates,1,3),[Y(end,:,:); CIMCVEC(:,:,1)],'--',...
    'LineWidth',2);
h3(1).Color = h1(1).Color;
h3(2).Color = h1(2).Color;
h3(3).Color = h1(3).Color;
h4 = plot(repmat(fDates,1,3),[Y(end,:,:); CIMCVEC(:,:,2)],'--',...
    'LineWidth',2);
h4(1).Color = h1(1).Color;
h4(2).Color = h1(2).Color;
h4(3).Color = h1(3).Color;
patch([fDates(1) fDates(1) fDates(end) fDates(end)],...
    [h2.YLim(1) h2.YLim(2) h2.YLim(2) h2.YLim(1)],'b','FaceAlpha',0.1)
xlabel('Year')
ylabel('Percent')
title('{\bf VEC Model Monte Carlo Forecasts}')
axis tight
grid on
legend(h1,DataTable.Properties.VariableNames(3:end),'Location','Best');
```



The plot suggests that `INT_S` has lower forecast accuracy that the other two series because its confidence bounds are the widest.

MMSE Forecasts Using VAR(p) Representation

Compute the autoregressive coefficient matrices of the equivalent VAR(3) model to the VEC(2) model (i.e., the coefficients of y_{t-1}, y_{t-2}, and y_{t-3}).

```
A = vec2var({B1 B2},C);
```

`A` is a 1-by-3 row cell vector. `A{j}` is the autoregressive coefficient matrix for lag term `j`. The constant offset (`b`) of the VEC(2) model and the constant offset of the equivalent VAR(3) model are equal.

Create a VAR(3) model object.

```
VAR3 = vgxset('AR',A,'a',b,'Q',EstCov);
```

Forecast over a 10 period horizon. Compute 95% individual, Wald-type confidence intervals for each series.

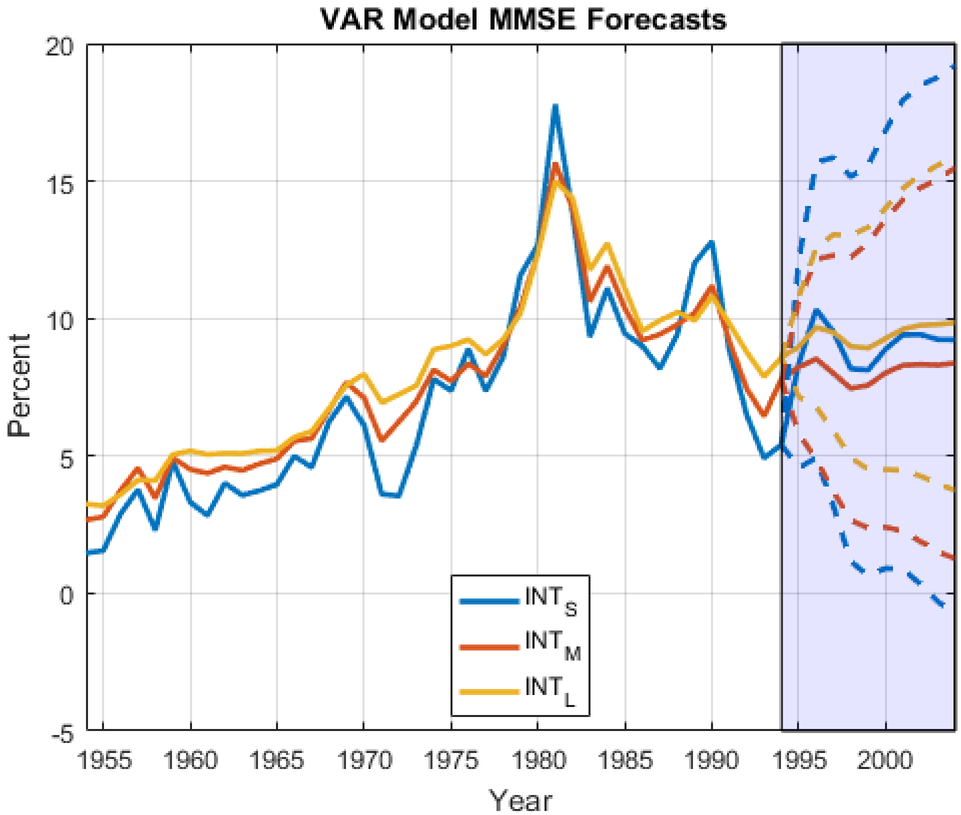
```
[MMSEF,CovF] = vgxpred(VAR3,forecastHorizon,[],Y);
var = cellfun(@diag,CovF,'UniformOutput',false);
CIF = zeros(forecastHorizon,numDims,2); % Preallocation
for j = 1:forecastHorizon
    stdev = sqrt(var{j});
    CIF(j,:,1) = MMSEF(j,:) - 1.96*stdev';
    CIF(j,:,2) = MMSEF(j,:) + 1.96*stdev';
end
```

The confidence intervals do not account for the correlation between the forecasted series at a particular time.

`MMSEF` is a 10-by-3 numeric matrix of the MMSE forecasts for the VAR(3) model. Rows correspond to forecast periods and columns to time series. `CovF` is a 10-by-1 cell vector of forecast covariance matrices, in which each row corresponds to a forecast period.

Plot the effective-sample observations, the MMSE forecasts, and the 95% Wald confidence intervals.

```
figure;
h1 = plot([dates; fDates(2:end)],[Y; MMSEF],'LineWidth',2);
h2 = gca;
hold on
h3 = plot(repmat(fDates,1,3),[Y(end,:,:); CIF(:,:,1)],'--',...
    'LineWidth',2);
h3(1).Color = h1(1).Color;
h3(2).Color = h1(2).Color;
h3(3).Color = h1(3).Color;
h4 = plot(repmat(fDates,1,3),[Y(end,:,:); CIF(:,:,2)],'--',...
    'LineWidth',2);
h4(1).Color = h1(1).Color;
h4(2).Color = h1(2).Color;
h4(3).Color = h1(3).Color;
patch([fDates(1) fDates(1) fDates(end) fDates(end)],...
    [h2.YLim(1) h2.YLim(2) h2.YLim(2) h2.YLim(1)],'b','FaceAlpha',0.1)
xlabel('Year')
ylabel('Percent')
title('{\bf VAR Model MMSE Forecasts}')
axis tight
grid on
legend(h1,DataTable.Properties.VariableNames(3:end),'Location','Best');
```



The MMSE forecasts are very close to the VEC(2) model Monte Carlo forecast means. However, the confidence bounds are wider.

Monte Carlo Forecasts Using VAR(p) Representation

Simulate 1000 paths of the VAR(3) model into the forecast horizon. Use the observations as presample data.

```
YSimVAR = vgxsim(VAR3,forecastHorizon,[],Y,[],numPaths);
```

`YSimVAR` is a 10-by-3-by-1000 numeric array similar to `YSimVEC`.

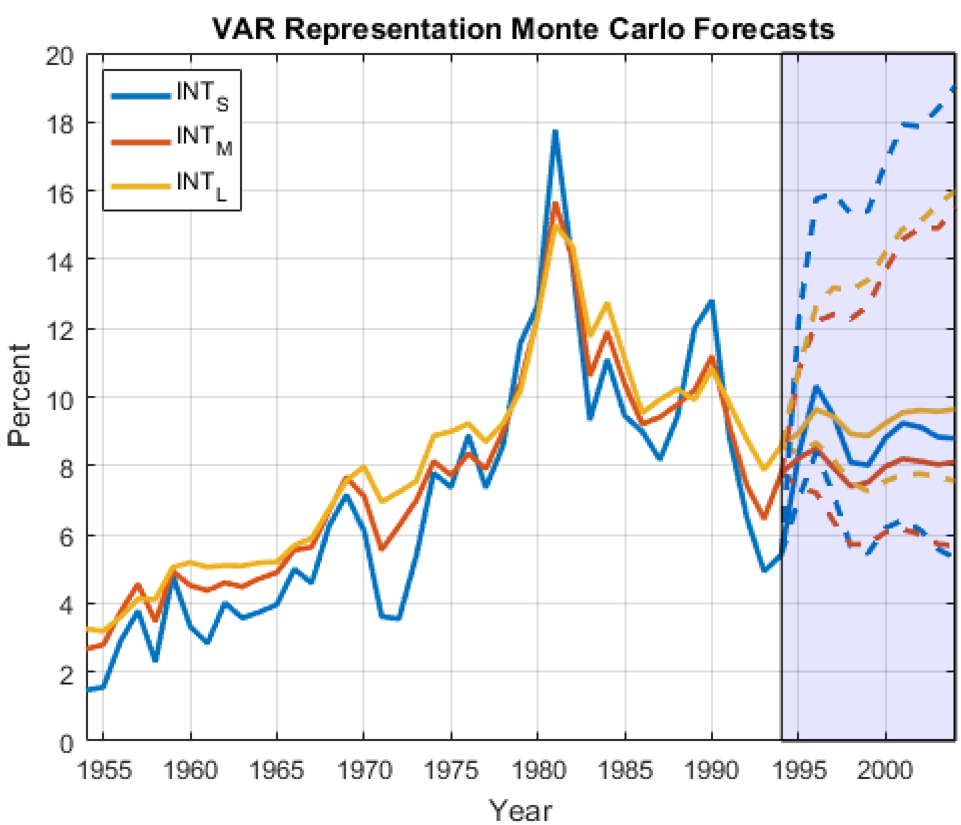
Compute the mean of the forecasts for each period and time series over all paths. Construct 95% percentile forecast intervals for each period and time series.

```
FMCVAR   = mean(YSimVAR,3);
CIMCVAR  = quantile(YSimVAR,[0.25,0.975],3);
```

`FMCVAR` is a 10-by-3 numeric matrix containing the Monte Carlo forecasts for each period and time series. `CIMCVAR` is a 10-by-3-by-2 numeric array containing the 2.5% and 97.5% percentiles of the draws for each period and time series.

Plot the effective-sample observations, the mean forecasts, and the 95% percentile confidence intervals.

```
figure;
h1 = plot([dates; fDates(2:end)],[Y; FMCVAR],'LineWidth',2);
h2 = gca;
hold on
h3 = plot(repmat(fDates,1,3),[Y(end,:,:); CIMCVAR(:,:,1)],'--',...
    'LineWidth',2);
h3(1).Color = h1(1).Color;
h3(2).Color = h1(2).Color;
h3(3).Color = h1(3).Color;
h4 = plot(repmat(fDates,1,3),[Y(end,:,:); CIMCVAR(:,:,2)],'--',...
    'LineWidth',2);
h4(1).Color = h1(1).Color;
h4(2).Color = h1(2).Color;
h4(3).Color = h1(3).Color;
patch([fDates(1) fDates(1) fDates(end) fDates(end)],...
    [h2.YLim(1) h2.YLim(2) h2.YLim(2) h2.YLim(1)],'b','FaceAlpha',0.1)
xlabel('Year')
ylabel('Percent')
title('{\bf VAR Representation Monte Carlo Forecasts}')
axis tight
grid on
legend(h1,DataTable.Properties.VariableNames(3:end),'Location','Best');
```



All sets of mean forecasts are very close. Both the VAR(3) and VEC(2) sets of Monte Carlo forecasts are almost equivalent.

See Also

egcitest | mvnrnd | vec2var | vgxpred | vgxset | vgxsim

Related Examples

- Test for Cointegration Using the Engle-Granger Test
- Estimate VEC Model Parameters Using egcitest

More About

- Vector Autoregressive (VAR) Models
- Cointegration and Error Correction Analysis
- Identifying Single Cointegrating Relations